Project Proposal

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This project investigates data-driven modeling and optimization of a gas–steam combined cycle power plant using deep learning methods. Combined-cycle units are complex thermo-mechanical systems characterized by strong nonlinearities, multi-variable couplings, and efficiency losses that vary significantly between steady and transient operating conditions. Traditional first-principles or control-oriented models struggle to capture these dynamics with high fidelity. Therefore, developing machine-learning-based surrogate models is both technically valuable and practically meaningful for improving operational flexibility, forecasting performance, and supporting optimization and diagnostic applications in modern power systems.

To provide context and background, the project will review literature on physics-based modeling (e.g., Rowen-type gas turbine models), empirical regression approaches for combined-cycle performance, hybrid frameworks such as physics-informed neural networks (PINNs), and recent applications of LSTM and Transformer architectures in industrial time-series prediction. This establishes the conceptual basis for selecting appropriate deep learning methods for different operating regimes.

The dataset consists of one year of operational measurements from an F-class gas–steam combined-cycle cogeneration plant. The data include temperatures, pressures, flows, turbine outputs, and steam cycle parameters at 10-minute resolution. Preprocessing will involve filtering, outlier removal, normalization, and segmentation into steady-state and transient subsets. For steady operating conditions, feedforward MLP models will be applied. For transient dynamics, sequence-based models such as LSTM and Transformer networks will be explored. Additionally, the project will investigate PINN-style constraints to embed thermo-physical knowledge into the learning process. Existing implementations from PyTorch and TensorFlow will be used and extended, with modifications to network structure, feature engineering, and loss functions.

Model performance will be evaluated through train/test prediction accuracy. Visualization will include time-series overlays of predicted vs. actual plant parameters, scatter plots, residual trends, and sensitivity analyses. Comparative evaluations across MLP, LSTM, Transformer, and PINN-inspired models will quantify the benefits and limitations of each method. Overall, the project aims to develop accurate, generalizable, and physically interpretable surrogate models for combined-cycle power plant analysis and optimization.